

COM3031-Probabilistic Machine Learning Workshop: Maximum A Posteriori

Overview

This workshop consists of two questions. In the first question, I provide a demonstration of the maximum a posteriori estimation of the mean and standard deviation of the normal distribution using one of the MCMC techniques via the PyMC library. Please, install the library if you opt to chose this for doing MCMC simulations. In the second question, you will estimate the posterior of the parameter in the Poisson distribution. You will also investigate the effect of prior parameters on the posterior. I will publish the solution in python next week. You may want to go through the lecture slides on the posterior distribution when you estimate the posterior analytically.

Maximum a Posteriori

1. Assume a data $\mathbf{y} = (y_1, \dots, y_n)^T$ is normally distributed.
 - (a) Write a script to get the mean and variance by maximising the likelihood.
 - (b) Assume the mean has its own distribution (also known as prior distribution). Assume $\mu \sim \text{Normal}(\mu_0, \sigma_0^2)$, estimate the posterior distributions, i.e. $p(\mu|\mathbf{y})$ using Markov Chain Monte Carlo. In Python, you can use PyMC library to approximate the posterior and can try different values for μ_0 and σ_0^2 . You can also write a script to estimate the posterior analytically with known variance.
2. Von Bortkiewicz counted the number of Prussian cavalrymen who were killed to death by a horse each year during 1875-1894. The data has been provided to you on ELE. This is the count data and a Poisson distribution can be used to model the data.

$$\mathbf{y} \sim \text{Poisson}(\lambda) \quad \lambda > 0$$

$$p(\mathbf{y}|\lambda) \propto \prod_{i=1}^n \lambda^{y_i} e^{-\lambda}$$

where λ is the parameter and also describes the mean in the distribution.

- (a) Write a script to estimate the parameter λ by maximising the likelihood.
- (b) Assume Von Bortkiewicz liked Bayesian statistics and wanted to get a distribution on λ . He put a prior distribution on λ :

$$\begin{aligned} \lambda &\sim \text{Gamma}(\alpha, \beta) \\ p(\lambda) &\propto \lambda^{\alpha-1} e^{-\beta\lambda} \end{aligned}$$

Write a script to estimate the posterior $p(\lambda|\mathbf{y})$ analytically or using MCMC.

- (c) Study the effect of prior parameters α and β on the posterior distribution. In Gamma distribution, the mean is $\frac{\alpha}{\beta}$ and the standard deviation is $\sqrt{\frac{\alpha}{\beta^2}}$. You can put an informative prior (e.g. with very small standard deviation) and see the effect on the posterior distribution of λ .